the planet back in. If the planet gets pushed inward, centrifugal forces win and push it back out. So this is where Miller's planet lives, in my interpretation of *Interstellar*.[31]

The Slowing of Time, and Tidal Gravity

Among all stable, circular orbits around Gargantua, the orbit of Miller's planet is the closest to the black hole. This means it's the orbit with the maximum slowing of time. Seven years on Earth is one hour on Miller's planet. Time flows sixty thousand times more slowly there than on Earth! This is what Christopher Nolan wanted for his movie.

But being so close to Gargantua, in my interpretation of the movie, Miller's planet is subjected to enormous tidal gravity, so enormous that Gargantua's tidal forces almost tear the planet apart (Chapter 6). Almost, but not quite. Instead, they simply deform the planet. Deform it greatly (Figure 17.3). It bulges strongly toward and away from Gargantua.